

```
In[1]:= Clear["Global`*"];
```

0. Readme

1. (x', y', z') are represented by $(x3, y3, z3)$.
2. $V_{\lambda\lambda}^s$, $V_{\phi\phi}^s$, and V_{rr}^s in Table B1 and Eq. (B3) are represented by SphericalV $\lambda\lambda$, SphericalV $\phi\phi$, and SphericalV rr .
3. V_{xx}^c , V_{yy}^c , and V_{zz}^c in Table B2 and Eq. (B4) are represented by CartesianV xx , CartesianV yy , and CartesianV zz .
4. The integral kernels are considered and the complete expressions should be presented as $G\rho \int_{\lambda_1}^{\lambda_2} \int_{\phi_1}^{\phi_2} \int_{u_1}^{u_2} \kappa(\text{Kernel}) d u' d \phi' d \lambda'$.

1. Basic settings

```
In[2]:= A = x3 * Sin[λ] - y3 * Cos[λ];
        B = x3 * Cos[λ] + y3 * Sin[λ];
```

```
In[4]:= l = Sqrt[r^2 + x3^2 + y3^2 + z3^2 - 2 * r * (z3 * Sin[φ] + B * Cos[φ])];
```

```
In[5]:= x = r * Cos[φ] * Cos[λ];
        y = r * Cos[φ] * Sin[λ];
        z = r * Sin[φ];
```

```
In[8]:= Δx = x - x3;
        Δy = y - y3;
        Δz = z - z3;
```

```
In[11]:= dx = -Sin[λ] * Δx + Cos[λ] * Δy;
        dy = -Sin[φ] * Cos[λ] * Δx - Sin[φ] * Sin[λ] * Δy + Cos[φ] * Δz;
        dz = Cos[φ] * Cos[λ] * Δx + Cos[φ] * Sin[λ] * Δy + Sin[φ] * Δz;
```

2. Expressions from Table B1 involved in Eq. (B3)

```
In[14]:= SphericalVλλ = (-2 * r^2 + x3^2 + y3^2 - 2 * z3^2 - 3 * (x3^2 - y3^2) * Cos[2 * λ] +
                        4 * r * z3 * Sin[φ] + 4 * r * B * Cos[φ] - 6 * x3 * y3 * Sin[2 * λ]) / (2 * l^5);
```

```
In[15]:= SphericalVφφ = (-r^2 - x3^2 - y3^2 - z3^2 + 3 * z3^2 * (Cos[φ])^2 + 2 * r * z3 * Sin[φ] +
                        2 * B * Cos[φ] * (r - 3 * z3 * Sin[φ]) + 3 * B^2 * (Sin[φ])^2) / l^5;
```

```
In[16]:= SphericalVrr = (2 * r^2 - x3^2 - y3^2 - z3^2 -
                        (4 * r - 3 * z3 * Sin[φ] - 3 * B * Cos[φ]) * (z3 * Sin[φ] + B * Cos[φ])) / l^5;
```

3. Expressions from Table B2 involved in Eq. (B4)

```
In[17]:= CartesianVxx = (3 * dx^2 - l^2) / l^5;
```

```
In[18]:= CartesianVyy = (3 * dy^2 - l^2) / l^5;
```

```
In[19]:= CartesianVzz = (3 * dz^2 - l^2) / l^5;
```

4. Test the correctness of Eqs. (B3) and (B4)

```
In[20]:= Laplace1 = FullSimplify[SphericalV $\lambda\lambda$  + SphericalV $\phi\phi$  + SphericalV $r r$ ]
```

```
Out[20]=  
0
```

```
In[21]:= Laplace2 = FullSimplify[CartesianV $x x$  + CartesianV $y y$  + CartesianV $z z$ ]
```

```
Out[21]=  
0
```

```
In[22]:= NotebookSave[EvaluationNotebook[]];
```